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Short communication

Escherichia coli hyperepidemic clone ST410-A harboring $bla_{CTX\text{-}M\text{-}15}$ isolated from fresh vegetables in a municipal market in Quito-Ecuador

David Ortega-Paredes[a,b], Pedro Barba[b], Santiago Mena-López[c], Nathaly Espinel[a], Jeannete Zurita[a,b,*]

[a] Pontificia Universidad Católica del Ecuador, Facultad de Medicina, Quito, Ecuador
[b] Unidad de Investigaciones en Biomedicina, Zurita & Zurita Laboratorios, Quito, Ecuador
[c] Pontificia Universidad Católica del Ecuador, Escuela de Ciencias Geográficas, Quito, Ecuador

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ABSTRACT

Dissemination of Extended spectrum β-lactamases (ESBL) Enterobacteriaceae is a major medical threat. Vegetables and fruits, which are usually consumed raw, are a very suitable pathway for the spread of these bacteria from farm-to-fork. However, limited information exists regarding resistant bacteria and epidemic clones that are disseminated in vegetables and tap water in South America. We processed a total of 90 samples in triplicate of nine typically consumed raw vegetables from a central municipal market, and tap water samples were processed from twenty-one locations in Quito, Ecuador. The samples were analyzed for total coliforms and ESBL Enterobacteriaceae contamination using the dilution filtration method. ESBL *Escherichia coli* isolates were phenotypically and genotypically characterized. The water was free of Enterobacteriaceae, but all the vegetables and fruits (except for blackberries) presented total coliform counts. Watercress had the highest load of total coliforms (3.3 × 10E4). ESBL *E. coli* was detected in alfalfa, leaf lettuce and parsley/cilantro samples. Alfalfa had the highest load of ESBL *E. coli*/total coliforms (1/3.3 × 10E2). We identified *E. coli* ST44-A and ST410-A harboring $bla_{CTX\text{-}M\text{-}15}$ downstream of *ISEcp1*. Alfalfa and parsley/cilantro were contaminated with hyperepidemic *E. coli* ST410-A, which was resistant to quinolones and harbored $bla_{CTX\text{-}M\text{-}15}$. For the first time, we report ESBL *E. coli* ST410-A from vegetables and express an alert regarding the health risk this could represent.

1. Introduction

Rapid dissemination of Enterobacteriaceae producing Extended-Spectrum β-lactamases (ESBL) is a major medical threat. This is mainly because these strains increase the therapeutic use of last-resort antimicrobials such as carbapenems, tigecycline and colistin. Therefore, antimicrobial resistance increases, and new mechanisms emerge (Iovleva and Doi, 2017; Srinivas and Rivard, 2017; Talaga-Ćwiertnia et al., 2017). The most frequent plasmid mediated ESBL include CTX-M, TEM, SHV and OXA types, and all of them are reported in clinical and environmental isolates of gram-negative bacilli. However, the CTX-M family is the most important enzymatic mechanism of resistance to third generation cephalosporins (3GC) in Enterobacteriaceae (Cantón et al., 2012).

To date, > 170 variants of $bla_{CTX\text{-}M}$ have been reported (http://www.lahey.org; last accessed October 2017) belonging to five phylogenetic groups (CTX-M-1, 2, 9, 8 and 25) based on their amino acid sequences (Poirel et al., 2008). Worldwide, $bla_{CTX\text{-}M\text{-}15}$, which is classified into the CTX-M-1 group, is most frequently reported in *Escherichia coli* isolated from clinical and environmental settings. The successful dissemination and establishment of this variant is related to its association with the pandemic clone known as ST131, transposition element IS*Ecp1* and its location in conjugative plasmids (Mathers et al., 2015).

Additionally, regarding the clinical location of *E. coli* producing CTX-M-15, an increase in reports of *E. coli* harboring the $bla_{CTX\text{-}M\text{-}15}$ gene isolated from the environment and food have exposed a complex model of dissemination and variety of reservoirs of this resistance mechanism (Randall et al., 2016; Röderova et al., 2017). The principal risk for public health is the evolution and dissemination of multidrug resistant hyperepidemic clones with capacity for spread from environment and animals to humans where cause difficult treatment infections (Müller et al., 2016; Schaufler et al., 2017). In this context, fresh vegetables are especially important because their production process, i.e., the contact with manure or sewage irrigation contaminated water, may lead to their contamination with resistant *E. coli* (Araújo et al., 2017).

* Corresponding author at: Pontificia Universidad Católica del Ecuador, Facultad de Medicina, Av. 12 de Octubre 1076 y Roca, Quito, P.O. Box: 17-01-2184, Ecuador.
E-mail address: jzurita@zuritalaboratorios.com (J. Zurita).

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Moreover, these products are commonly consumed raw, which results in a very suitable vehicle for human gut colonization [9].

To establish the contribution of vegetables and fruits in the dissemination of ESBL Enterobacteriaceae, identification of the pathways of bacterial spread to humans from farm-to-fork is needed. However, few reports of ESBL coliforms in raw vegetables exist, and to the best of our knowledge, there are no reports from South America in the indexed literature. Therefore, the aim of this pilot study was to detect ESBL coliforms on fresh vegetables commercialized at a central municipal market (CMM) in Quito, Ecuador.

2. Materials and methods

2.1. Study design

Quito is in the Andean Region of Ecuador (0°13′07″S and 78°30′35″W) and located at an altitude of 2810 m. Agricultural products that supply the city come mostly from the surrounding valleys and highlands and from the coastal region of the country. Because municipal markets are the main distribution system of vegetables and fruits for retail stores, we decided to sample from a CMM in Quito. To detect coliform contamination, we selected nine fresh vegetables and fruits that are mostly consumed raw in salads and juices: iceberg lettuce, leaf lettuce, watercress, alfalfa, parsley/cilantro, green pepper, tomato, strawberry, and blackberry. Additionally, tap water is the main source of water used to wash and prepare food. Therefore, we considered the coliform evaluation of tap water for possible source of contamination.

2.2. Sampling

During samples acquisition, we also asked sellers about the farming source of the products. Samplings of vegetables and fruits were conducted at three-time periods: June, July and August 2015. We collected approximately 500 g of vegetables and fruits three times each product (in the first week of each sampling months) to a total of 27 samples and triplicate water samples from twenty-one locations around the city were obtained during August 2015. A detailed map showing the selected sampling sites was constructed in ArcGIS software using waypoints registered with a GPS eTrex 10 (Garmin) over a set of Pleiades high-resolution satellite images (ortho, RGB, 2012–2013, and pixel 0.5 × 0.5 m); another complementary map shows the general location and was generated using Google Maps with open data (https://www.google.com.ec/maps) (Fig. 1).

2.3. Quantification and phenotypic characterization

Vegetable and fruit samples were acquired and immediately packaged aseptically in sterile polythene zip bags, and water samples were collected in sterile plastic flasks. The samples were transported in a cold box for analysis within 2–3 h. Forty grams of the edible parts of each sample were selected to simulate salad or juice preparation; the samples were not washed, and the inedible or damaged parts were removed using a sterile knife. The selected parts were suspended in 460 mL of peptone water and shaken for 30 min at room temperature before coliform analysis.

Quantitative evaluation of total and 3GC resistant coliforms was conducted by a filtration method on McConkey plates. Briefly, 10 µL, 100 µL, 1 mL and 10 mL of each sample was mixed with 100 mL of saline water and filtered through a 50 µm filter (Millipore). The membranes were deposited on McConkey plates without antimicrobial selection for counts of total coliforms and deposited on plates supplemented with cefotaxime (5 µg/mL) for isolation of ESBL coliforms. The plates were incubated at 37 °C for 24 h. Enumeration was manually conducted. All coliform-like colonies were enumerated to establish total and 3GC resistant coliform loads. Fisher's exact test was used to identify

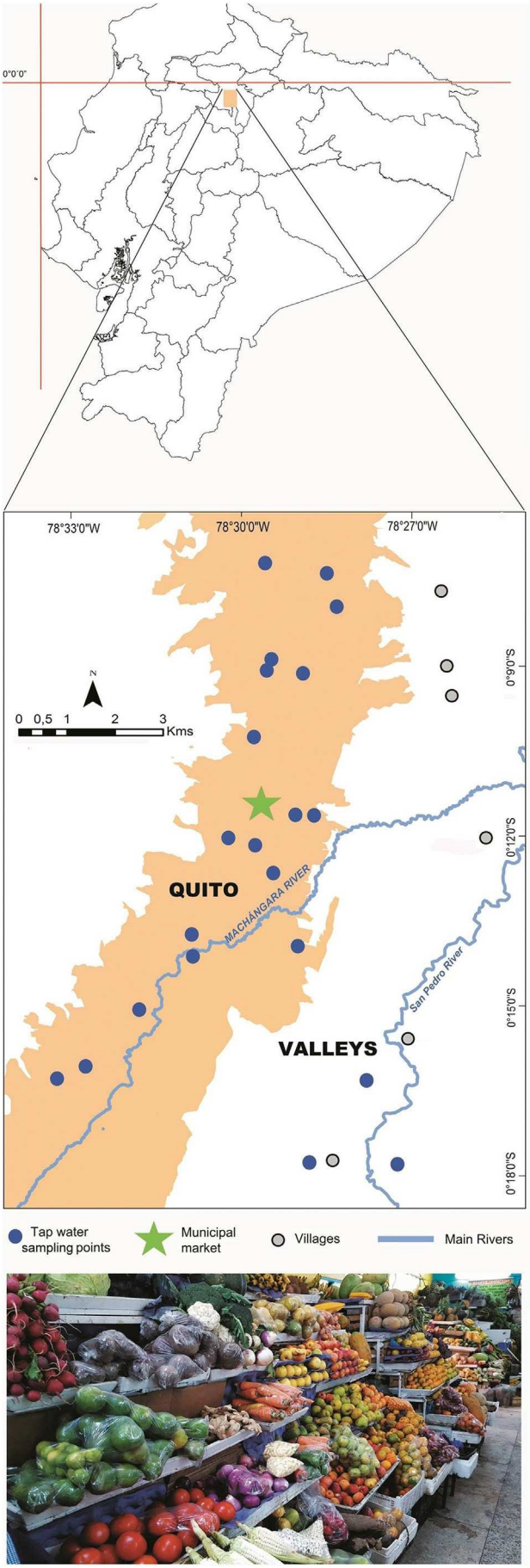


Fig. 1. Study area and location of sampling points.

significant differences between samplings using RStudio free version software. All coliform-like colonies on cefotaxime supplemented plates and different coliform colonies on total coliform plates were spiked, and species identification was completed. Tap water samples (100 mL) were analyzed by applying the same method.

The species was identified by a Microflex LT MALDI-TOF spectrometer (Bruker Daltonics Inc.) and examined in the MALDI-TOF Biotyper v. 3.1. One 3GC resistant *E. coli* isolate per positive sample was randomly selected for detection of ESBL production using double-disk test and MIC profiles were performed using a VITEK®2 Compact (bioMérieux, USA). Susceptibility profiles were established following the CLSI, 2016 guidelines.

2.4. Genotypic characterization of ESBL E. coli

Phenotypically characterized ESBL *E. coli* were screened for $bla_{\text{CTX-M group 1}}$, $bla_{\text{CTX-M group 2}}$, $bla_{\text{CTX-M group 9}}$, bla_{KPC}, *mcr-1* genes and the integron 1 variable region were amplified and sequenced using a dideoxy sequencing method as previously conducted (Ortega-Paredes et al., 2016; Zurita et al., 2016). The location of IS*Ecp1* was achieved by PCR mapping (Zurita et al., 2016). Typing was conducted using Rep-PCR (Araújo et al., 2017), and MLST was achieved using a scheme proposed by the University of Warwick (Wirth et al.; 2006). Phylogroups were assigned as described elsewhere (Clermont et al., 2000). Our own positive and negative controls were included in each experiment.

3. Results

3.1. Sampling

In this study, the precise production site of the sampled vegetable was hard to determine. However, data from sellers revealed that iceberg lettuce, leaf lettuce, watercress, alfalfa, parsley/cilantro, tomatoes, strawberries, and blackberries were from the highlands, and green peppers were from the coastal region.

3.2. Phenotypic characterization and quantification

All vegetable and fruit samples were positive for total coliforms in the three collections except for blackberries, which were negative in the three samplings. The average of the total coliforms enumeration ranged from 50 CFU/g in tomatoes to 3.3 × 10E4 in watercress. On the other hand, 3GC resistant coliforms were detected only in leaf lettuce, alfalfa and parsley/cilantro (Table 1). No significant differences between samplings were detected (> 0.005) and none of the tap water samples were positive for the total or ESBL coliforms analysis.

MALDI-TOF analysis identified as *E. coli* all coliform-like colonies (65) spiked from cefotaxime supplemented plates and different coliform colonies (25) were identified as *Citrobacter freundii* (2), *C. koseri* (3), *Enterobacter amnigenus* (1), *E. asburiae* (1), *E. cloacae* (3), *E. gergoviae* (1), *Erwinia persicina* (1), *Escherichia coli* (3), *E. vulneris* (1), *Ewingella americana* (1), *Klebsiella oxytoca* (3), *Leclercia adecarboxylata* (2), *Pantoea agglomerans* (2) and *P. ananatis* (1).

The rate between ESBL *E. coli*/total coliforms was higher in alfalfa (1/3.3 × 10E2), leaf lettuce (1/4.6 × 10E2) and parsley/cilantro (1/5 × 10E2) samples. Furthermore, alfalfa samples were positive for ESBL *E. coli* in the three samplings, whereas leaf lettuce and parsley/cilantro were positive in the first and last samplings (Table 1). Seven *E. coli* (one per positive sample) were positive for ESBL-production by double-disk test and were multidrug resistant, but none of them were resistant to carbapenems and cefoxitin (Fig. 2).

3.3. Genotypic characterization of ESBL E. coli

Rep-PCR analysis classified these isolates into two groups, and MLST recognized them as ST410 (two isolates) and ST44 (five isolates). Interestingly, ST410 was found only in the first sampling, whereas ST44 was identified in all the samplings. Alfalfa and parsley/cilantro were contaminated with two sequence types (ST410 in the first sampling and ST44 in the second and last samplings). Three isolates, which belonged to ST44 (two isolates) and ST410 (one isolate), presented the same integron 1 variable region (*dfrA1* and *addA5*). All seven isolates presented $bla_{\text{CTX-M-15}}$ downstream of *ISEcp1* and belonged to phylogroup A (Fig. 2).

4. Discussion

Vegetables and fruits normally carry non-pathogenic epiphytic bacteria. However, the use of manure for fertilizing or sewage for irrigation leads to contamination with pathogenic bacteria, particularly those belonging to the Enterobacteriaceae family. In our location, these production practices are common, and vegetables and fruits are consumed raw before washing with tap water. However, there is a lack of data regarding fresh products and tap water contamination related to antibiotic resistance and the spread of epidemic clones.

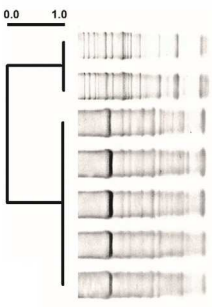
Even though our results reveal higher total coliform loads in watercress (3.3 × 10E4), ESBL *E. coli* was isolated only in leaf lettuce, alfalfa and parsley/cilantro. This issue could be due to the production system. While watercress is an aquatic plant mostly collected from natural rivers, vegetables contaminated with ESBL *E. coli* mostly come from intensive agricultural fields where contamination with resistant bacteria from animal and human sources could occur during growth, harvest or transportation. To confirm this observation, microbial source tracking of the contamination of these vegetables is necessary.

In contrast, there is no evidence of coliform contamination of tap water in the city. Therefore, washing vegetables and fruits using tap water from Quito could block the route of dissemination before ESBL *E. coli* reaches consumers. However, this practice is not always the rule and mainly occurs with street and restaurant food. Additionally, alfalfa, which is used for traditional juice preparation, is also used to feed small herbivores (principally guinea pigs and rabbits) at production farms,

Table 1
Average quantification of total coliforms and ESBL *E. coli* on samples.

Sample	Total coliforms (CFU/g)[a]	ESBL *E. coli* coliforms (CFU/g)[a]	Ratio ESBL *E. coli*/ total coliform (CFU/g)
Watercress	3.3 × 10E4	–	–
Leaf lettuce	2.3 × 10E3	5	1/4.6 × 10E2
Iceberg lettuce	2.1 × 10E3	–	–
Parsley/cilantro	2 × 10E3	4	1/5 × 10E2
Alfalfa	5 × 10E2	15	1/3.3 × 10E2
Strawberry	4 × 10E2	–	–
Green pepper	2 × 10E2	–	–
Tomato	50	–	–

[a] Data shown are the average of positive samples counts.



Isolate	Vegetable	Date	ST	Phylogroup	Group	ESBL	INT1 RV	ISEcp1	AMP	SAM	TPZ	CZ	FOX	CAZ	CRO	FEP	ERT	IMI	GN	TBR	CIP	LEV	NIT	SXT
PSC 03	parsley/cilantro	02/06/15	410	A	I	$bla_{CTX-M-15}$	–	+	≥32	≥32	≥128	≥64	≤4	16	≥64	≥64	≤0.5	≤0.25	≤1	≥16	≥4	≥8	≤16	≤20
PSC 07	Alfalfa	02/06/15	410	A	I	$bla_{CTX-M-15}$	*dfrA1, aadA5*	+	≥32	≥32	≥128	≥64	≤4	16	≥64	≥64	≤0.5	≤0.25	≤1	≥16	≥4	≥8	≤16	≥320
PSC 09	Leaf lettuce	02/06/15	44	A	II	$bla_{CTX-M-15}$	*dfrA1, aadA5*	+	≥32	≥32	8	≥64	≤4	16	≥64	2	≤0.5	≤0.25	≥16	8	≥4	≥8	≥64	≥320
PSC 18	Alfalfa	05/07/15	44	A	II	$bla_{CTX-M-15}$	–	+	≥32	≥32	8	≥64	=4	16	≥64	2	≤0.5	≤0.25	≥16	8	≥4	≥8	≥64	≥320
PSC 19	parsley/cilantro	02/08/15	44	A	II	$bla_{CTX-M-15}$	*dfrA1, aadA5*	+	≥32	≥32	8	≥64	≤4	16	≥64	2	≤0.5	≤0.25	≤1	8	≥4	≥8	≤16	≥320
PSC 20	Leaf lettuce	02/08/15	44	A	II	$bla_{CTX-M-15}$	–	+	≥32	≥32	8	≥64	≤4	16	≥64	2	≤0.5	≤0.25	≤1	8	≥4	≥8	≤16	≥320
PSC 21	Alfalfa	02/08/15	44	A	II	$bla_{CTX-M-15}$	–	+	≥32	≥32	8	≥64	≤4	16	≥64	2	≤0.5	≤0.25	≤1	8	≥4	≥8	≤16	≥320

Fig. 2. Antibiotic sensitivity profile and genotypic characterization of the isolates.

backyard farms and households. Feeding animals with products contaminated with ESBL *E. coli* can increase the dissemination of multidrug-resistant bacteria and cause horizontal transfer of resistance among bacterial populations.

Occurrence of raw vegetables contaminated with ESBL *E. coli* has been reported elsewhere (van Hoek et al., 2015; Zurfluh et al.; 2015), but the isolates generally belong to STs related with the environment rather than human sources or with a capacity for interspecific dissemination. However, ESBL *E. coli* epidemic clones such as 131, 405 and 410 have been isolated in the animal food production chain, wastewater and humans (Müller et al., 2016; Silva et al., 2016).

In this study, we report on *E. coli* ST44 and ST410. While ST44 is a clone related with multidrug resistance, especially resistance to quinolones, ST44 is not identified as an epidemic clone (Báez et al., 2015). However, *E. coli* ST410, which is the founder of the disseminated clonal complex 23 (CC23), has been identified as a new pandemic clone with interspecific dissemination capacity (Schaufler et al., 2016). Furthermore, *E. coli* ST410 has been found to harbor resistance genes such as KPC (Mavroidi et al., 2012), NDM (Qin et al., 2016) and the combination *mcr-1*/$bla_{CTX-M-15}$/resistant to quinolones, which was determined in clinical sampling (Rocha et al., 2017).

Clinical and environmental *E. coli* strains mainly fall into phylogenetic groups A, B1, B2, and D. *E. coli* phylogenetic groups A and B have been associated with the animal commensal origin and a higher level of resistance than B2 and D (Chakraborty et al., 2015). All isolates in our study belong to group A, which suggests a possible fecal contamination of vegetables. Additionally, three isolates presented aminoglycoside resistance genes located in the integron 1 variable region. The presence of this genetic element allows these strains to recombine more cassette genes of resistance and increase their resistance levels.

5. Conclusions

This is the first identification of this hyperepidemic clone, ST410 ESBL, which is resistant to ciprofloxacin and belongs to the phylogenetic group A (chuA/yjaA/TSpE4.C2: −+−) (Rocha et al., 2017) in fresh vegetables. These findings emphasize the necessity to pay close attention to this clone as a probable future public health Threat. Finally, it is necessary to deepen and extend this study using cost effective techniques to establish the prevalence and pathways by epidemic multidrug-resistant clones of *E. coli* that can spread to humans from farm-to-fork.

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Conflicts of interest

None.

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